

Employee Payroll Management

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Acknowledgement

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With Sincere Regards,

Hiya Bhaveshkumar Patel

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Project Profile

Project Profile

Project Name: Employee Payroll Management

Type of Application: Desktop Application

Application:

Project Description: This project is an Employee Payroll Management System developed using Python (Tkinter) and SQLite. It allows users to manage employee records, calculate salaries automatically, and generate salary receipts. The system ensures accurate payroll processing through validations and a user-friendly graphical interface.

Team Size: 1

Front End: Python

Back End: SQLite3 Database

Introduction to Tools

Introduction to Tools

Front-End Tools (Tkinter GUI App)

1. Tkinter (Standard Python GUI)

- **Widgets Used:**
 - Tk(): Initializes the main application window.
 - Label(): Displays headings and text.
 - Entry(): Single-line text input (employee details, salary fields).
 - Button(): Perform actions (Save, Update, Delete, Calculate, Search).
 - Frame(): Organizes the GUI into sections
 - messagebox: Displays error, warning, and success messages

2. ttk (Themed Tkinter Widgets)

- **Used for modern-looking widgets:**
 - ttk.Combobox – Dropdown selections (Gender, Status, Month, Year, Proof ID)
 - ttk.Treeview – Displays employee records in tabular format
 - Scrollbar – Vertical and horizontal scrolling

3. tkcalendar

- **Used for date selection:**
 - DateEntry – Calendar-based input for Date of Birth (DOB) and Date of Joining (DOJ).

4. Layout & Styling

- Layout Managers: place() and pack()
- Styling Options: bg, fg, font, cursor, relief, bd

5. Dynamic Updates & Validation

- Automatic **age calculation** from DOB
- **Total days calculation** based on selected month and year
- Input validation using StringVar(), regex, and key validation

6. Multiple Windows

- Toplevel() – Opens a separate employee details window
- Custom window size using geometry()

7. User Interaction

- bind() – Double-click event on Treeview to open salary receipt
- Real-time input restrictions (contact number validation)

8. Variables

- **StringVar()** – Used to link GUI inputs with backend logic

Back-End Tool: SQLite (Using sqlite3)

Purpose:

- Manages employee and payroll data in a local database(emp.db).
- Stores employee personal details, salary details, and salary receipt file paths.
- Enables secure, offline data storage for the desktop application.

Key Operations:

1. Connect to DB

- `con = sqlite3.connect('emp.db')`
- `cur = con.cursor()`
- Establishes a connection with the SQLite database.

2. Insert Data

- Inserts employee and salary details into the e_salary table.
- Uses parameterized queries (?) to prevent SQL injection.
- `con.commit()` saves changes permanently.
- `messagebox.showinfo()` confirms successful insertion.

3. Retrieve Data

- `SELECT * FROM e_salary`
- Fetches all employee records.
- Displays data in `ttk.Treeview` for tabular viewing.

4. Search Record

- `SELECT * FROM e_salary WHERE emp_code=?`
- Searches employee data using Employee Code.
- Displays record or error message if not found.

5. Update Data

- UPDATE e_salary SET ... WHERE emp_code=?
- Updates employee and salary details after validation.
- Recalculates net salary before saving.

6. Delete Data

- DELETE FROM e_salary WHERE emp_code=?
- Deletes employee records from the database.
- Confirms record existence before deletion.

Functions Using SQLite

- add(): Insert new employee & payroll data
- search() – Retrieve employee record by Employee Code
- show() – Display all employee records
- update() – Modify existing employee details
- delete() – Remove employee record
- check_connection() – Verify database connectivity

❖ Supporting Tools & Features (Backend & Utility)

❖ MessageBox (tkinter.messagebox)

- Purpose: Displays alerts, warnings, and confirmation messages.
- Used for:
 - Input validation errors
 - Database success/failure messages
 - Delete and update confirmations

❖ Time Module (time)

- Purpose: Shows real-time clock and auto-refresh functionality in the GUI.
- Functions:
 - strftime(): Formats and displays current date and time.
 - after(200, ...): Updates time and UI elements periodically.

❖ OS Module (os)

- Purpose: Handles system-level file and directory operations.
- Used for:
 - Opening salary receipt files
 - Checking file paths
 - Managing invoice/receipt storage

❖ Calendar Tool (tkcalendar)

- Purpose: Allows date selection through a calendar widget.
- Used for:
 - Salary date / joining date selection
- Widget:
 - DateEntry

❖ Regular Expressions (re)

- **Purpose:** Validates user input formats.
- **Used for:**
 - Mobile number validation
 - Email format validation

❖ Math Calculations

- **Purpose:** Salary computation logic.
- Used for:
 - Gross salary calculation
 - Deductions (PF, tax, allowances)
 - Net salary calculation

❖ File Handling

- **Purpose:** Saves and retrieves salary receipts.
- Used for:
 - Writing payroll data to text/PDF files
 - Loading saved receipts

System Study

Existing System

1. Employee Management

- Stores employee and professional details.
- Displays all employee records in tabular format.

2. Payroll Management

- Manages employee salary details such as basic salary, working days, absences, PF, medical, and conveyance.
- Automatically calculates net salary.

3. Attendance-Based Salary Calculation

- Calculates salary based on total days, present days, and absent days.
- Prevents invalid attendance entries.

4. Salary Receipt Management

- Generates salary slips for each employee.
- Stores salary receipts as text files in a local folder.

5. Search & Record Tracking

- Searches employee records using Employee Code.
- Allows viewing past salary records.

6. Update & Delete Records

- Updates existing employee and payroll details.
- Deletes employee records securely with confirmation.

7. Validation & Error Handling

- Validates email, contact number, dates, and numeric fields..
- Displays error and confirmation messages using dialog boxes.

8. Desktop-Based Operation

- Runs as a standalone desktop application.
- Uses a local SQLite database without internet dependency.

Proposed System

1. Improved User Interface (UI)

- User-friendly and structured interface developed using Tkinter GUI.
- Clear form layout with labeled input fields for employee and payroll details.
- Consistent color scheme and sectioned frames for better usability.

2. Automated Salary Processing

- Automatic calculation of gross salary, deductions, and net salary.
- Attendance-based salary computation reduces manual errors
- Instant salary updates upon data entry or modification.

3. Real-Time Updates

- Employee payroll records update immediately in the data table.
- Live date and time display using system clock.
- Dynamic refresh of salary calculations and records.

4. Modular and Scalable System

- Separate modules for:
 - Employee details
 - Payroll calculation
 - Salary receipt generation
- Easy to extend with future features like reports, tax calculation, and export options.

5. Improved Database Management (SQLite3)

- Uses **SQLite database (emp.db)** for structured data storage.
- Secure SQL queries with parameter binding (?) to prevent SQL injection.
- Centralized database handling for smooth data operations.

6. Salary Receipt Management

- Automatically generates salary slips for employees.
- Stores salary receipts in a local directory for future reference.
- Enables easy access to previous payroll records.

7. User Authentication & Logout System

- Scope for future role-based access (Admin, HR, Accountant).

8. Error Handling and Exception Management

- Uses try-except blocks to handle runtime and database errors.
- Displays user-friendly error messages using message boxes.
- Prevents system crashes during data processing.

Scope of the Project

❖ Functional Scope

1. User Management

- Secure login and logout functionality for system access.
- Scope for future role-based access (Admin, HR, Accountant).
- Controlled access to employee payroll data.

2. Employee Management

- Add, update, and delete employee information.
- Stores employee information such as name, contact number, email, designation, and employee code.
- Displays all employee records in a tabular format.

3. Salary Receipt Generation

- Generates salary slips for employees.
- Stores salary receipts in a local directory for future reference.
- Allows retrieval and viewing of previous salary records.

4. Payroll Management

- Manages salary components including basic salary, allowances, deductions, and bonuses.
- Attendance-based salary calculation.
- Automatic computation of gross salary and net salary.

5. Search and Record Tracking

- Search employee payroll records using employee code.
- Quick access to specific employee salary details

6. Validation & Error Handling

- Validates employee inputs such as email, mobile number, and salary fields.
- Displays warnings and confirmations using dialog boxes.

7. Dashboard & Real-Time Updates

- Displays employee payroll records dynamically.
- Automatic refresh of payroll data after updates.

❖ Technical Scope

1. Technology Used

- **Python** – Backend logic and application control.
- **Tkinter** – GUI development.
- **SQLite3** – Local database management.
- **Pillow (PIL)** – Image handling in GUI.
- **tkcalendar** – Date selection feature.

2. Modular Design

- Uses separate functions and classes for Employee handling, Payroll calculation, Database operations.
- Ensures easy maintenance and scalability.

3. Database Management

- Stores employee and payroll data in emp.db.
- Uses parameterized SQL queries for secure operations.
- Efficient retrieval, insertion, update, and deletion of records.

4. Error Handling

- Uses try-except blocks to handle runtime and database errors.
- Prevents application crashes.
- Displays user-friendly error messages using Tkinter messagebox.

Aim& Objective of the project

❖ Aim:

- The Employee Payroll Management System (EPM) aims to automate and simplify the process of managing employee records and salary calculations through a secure, accurate, and user-friendly desktop application.
- The system is designed to reduce manual payroll errors, improve efficiency, and ensure reliable salary processing using an integrated database.

❖ Objectives:

1. Efficient Employee Management

- Maintain complete employee records including personal and job-related details.
- Enable easy addition, updating, searching, and deletion of employee data.
- Ensure organized and centralized employee information storage.

2. Automated Payroll Processing

- Automatically calculate employee salaries based on attendance and salary components.
- Reduce human errors in salary computation.
- Generate accurate net salary values instantly.

3. Salary Slip Generation & Storage

- Generate salary receipts for each employee.
- Store salary slips securely in a local directory for future reference.
- Allow retrieval of previous payroll records.

4. Real-Time Data Updates

- Display employee payroll data dynamically in tabular format.
- Automatically update salary records after modifications
- Show live date and time on the application interface.

5. Secure & User-Friendly System

- Implement secure login and logout functionality
- Provide an easy-to-use graphical interface using Tkinter.
- Ensure data integrity and consistency using an SQLite database.

6. Error Handling & Validation

- Validate user inputs such as contact number, email, and salary fields.
- Prevent incorrect data entry using validation checks.
- Display meaningful error and confirmation messages.

Feasibility Study

- A feasibility study is conducted to evaluate the practicality of developing a new system or improving an existing payroll process.
- The main objective is to analyze whether the proposed Employee Payroll Management System is technically, economically, and operationally feasible.
- This study helps in understanding project viability and avoids unnecessary investment of time and resources.

1. Technical Feasibility

- Determines whether the system can be developed using available technology and skills.
- The project uses Python with Tkinter for creating a user-friendly desktop interface.
- SQLite is used as a lightweight and efficient database for payroll data storage.
- Supports real-time data updates such as salary calculations and record refresh.
- Runs smoothly on standard systems without requiring high-end hardware.
- Hence, the project is technically feasible due to its use of reliable and widely available technologies.

2. Economic Feasibility (Cost-Benefit Analysis)

- Evaluates whether the project is cost-effective.
- Development Cost: Very low, as all tools used (Python, Tkinter, SQLite) are open-source.
- Maintenance Cost: Minimal, requiring basic database management.
- Labor Cost: No need for specialized technical staff due to simple operation.
- Benefits: Reduces manual payroll errors, saves time, and improves salary accuracy.
- Therefore, the system offers high benefits with minimal cost, making it economically feasible.

3. Operational Feasibility

- Assesses whether the system can be easily integrated into existing payroll operations.
- Provides a simple and user-friendly interface with clear input fields and buttons.
- Automates manual payroll tasks such as salary calculation and record maintenance.
- Reduces human errors and improves efficiency in payroll processing.
- Requires minimal user training, making it operationally feasible.

System Analysis

Requirement Specification

Project Module

1. Admin Module

❖ **Role:**

- The Admin has full control over the payroll system..
- Responsible for managing employee records, payroll details, and salary reports.

❖ **Features:**

- Dashboard – View all employee payroll records in tabular form.
- Manage Employees – Add, update, search, and delete employee details.
- Payroll Management – Enter salary components such as basic salary, allowances, deductions, and attendance.
- Salary Calculation – Automatically calculate gross salary and net salary.
- Salary Slip Generation – Generate and store salary receipts.
- Search Records – Retrieve employee payroll details using Employee Code..
- Billing System – Generate customer invoices.
- Logout System – Secure login/logout functionality.

❖ **Files Used:**

- employee.py – Main payroll and employee management system

2. Employee Module

❖ **Role:**

- Employees can view their own payroll information (future enhancement).
- No permission to modify salary or employee records.

❖ **Features:**

- View Salary Details – View personal salary slip.
- View Attendance-Based Salary – Check present/absent days and salary calculation.
- Limited Access– Cannot edit or delete records.

❖ **Files Used:**

- employee.py (Read-only access – future scope)

3. Payroll Database Module

❖ Role:

- Handles all backend database operations related to employees and payroll.

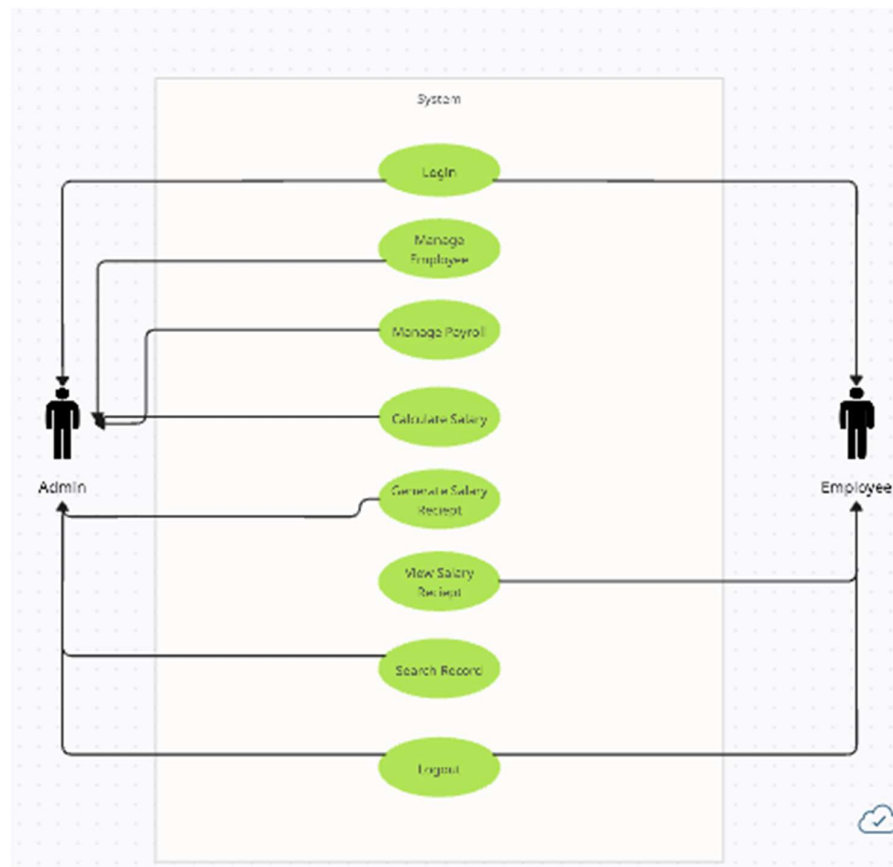
❖ Features:

- Stores Data Securely – Uses SQLite database (emp.db)
- Manages Payroll Records – Saves employee details, salary data, and attendance.
- Supports CRUD Operations– Create, Read, Update, Delete payroll records.
- Ensures Data Integrity – Uses parameterized SQL queries.

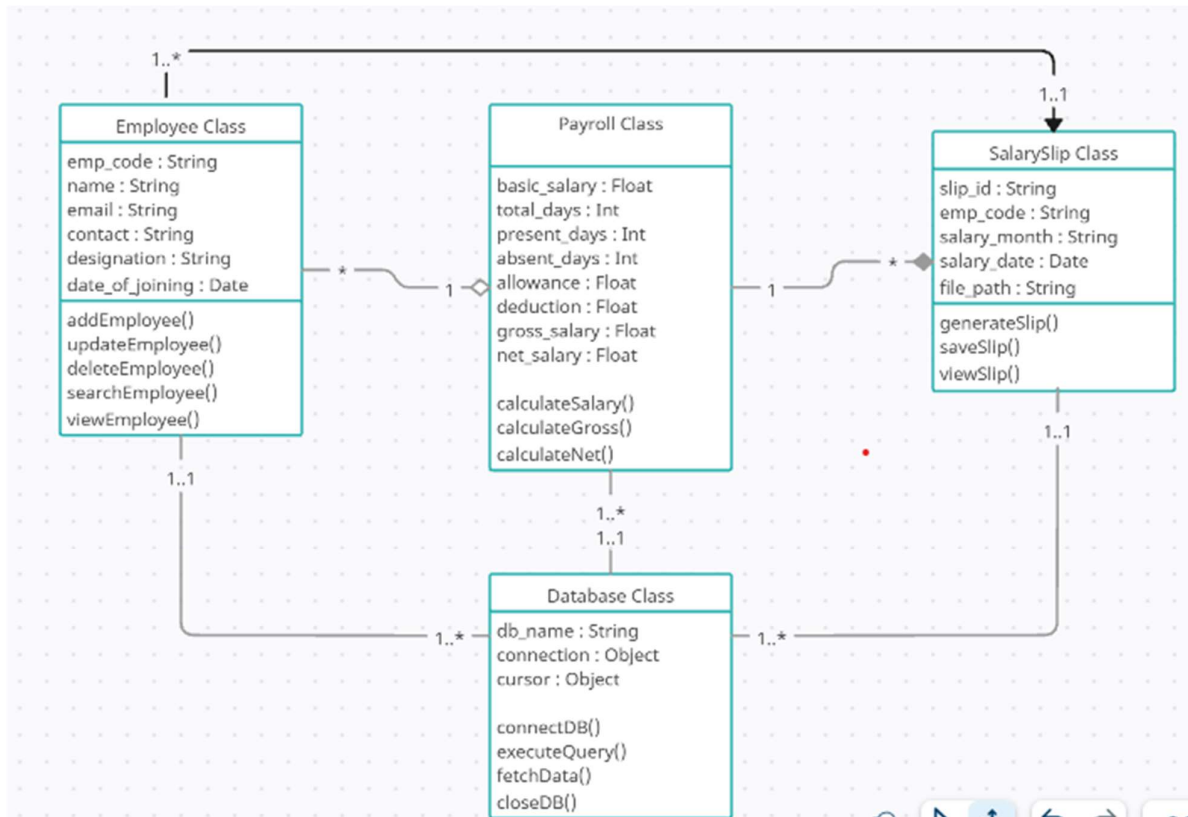
❖ Files Used:

- emp.db – SQLite database file
- sqlite3 module – Database connectivity

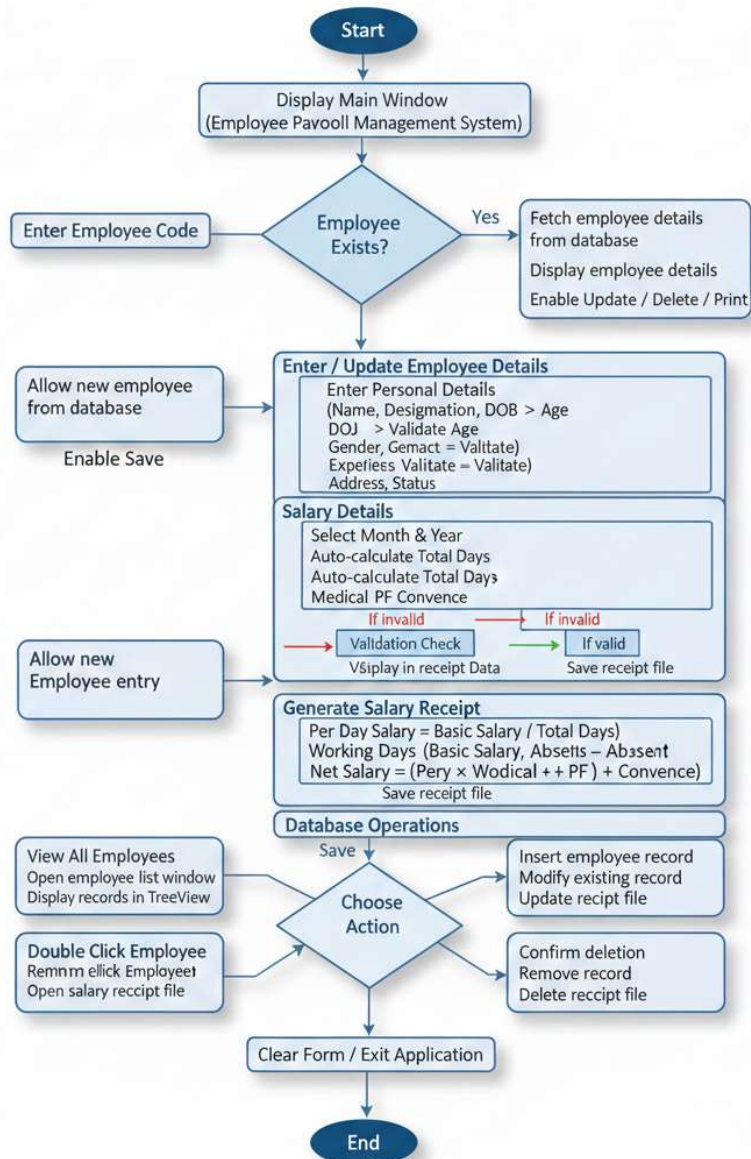
Use Case Diagram



Class Diagram



Activity Diagram



System Design

Data Dictionary:**Table: e_salary**

Name	Type	Constraints
emp_code	INT	Unique Employee Code (Primary Key)
designation	TEXT	Employee job designation
name	TEXT	Employee full name
age	INT	Employee age (auto-calculated from DOB)
gender	TEXT	Gender (Male/Female/Other)
email	TEXT	Employee email address
hr_location	TEXT	Hiring location / HR location
dob	TEXT(DATE)	Date of Birth (dd-mm-yyyy)
doj	TEXT(DATE)	Date of Joining (dd-mm-yyyy)
proof_id	TEXT	Identity proof type (Aadhar/PAN/License)
contact	TEXT	Contact number (10 digits)
status	TEXT	Employee status (Active/Inactive)
experience	INT	Total years of experience
address	TEXT	Residential address
month	TEXT	Salary month (Jan–Dec)
year	TEXT	Salary year
basic_salary	REAL	Basic monthly salary
total_days	INT	Total days in selected month
absent	INT	Number of absent days
medical	REAL	Medical deduction amount
pf	REAL	Provident Fund deduction
conveyance	REAL	Conveyance allowance
net_salary	REAL	Final calculated net salary
salary_receipt	TEXT	Path of saved salary receipt file

Screen Layout

Dashboard Page:

Employee Payroll Management System

EMPLOYEE PAYROLL MANAGEMENT SYSTEM

Employee

Employee Details

Employee Code:

Designation: D.O.B.:

Name: D.O.J.:

Age: Experience:

Gender: Proof ID:

Email: Contact:

Hired: Status:

Address:

Employee Salary Details

Month: Year: Basic Salary:

Total Days: Absent:

Medical: PF:

Convenience: Net Salary:

7	8	9	+
4	5	6	-
1	2	3	*
0	c	=	/

Salary Receipt

Company Name, TechSolution
Address: TechSolution, Floor4

Employee Code :
Employee Name :
Salary Of : Mon-YYYY
Generated On : DD-MM-YYYY

Total Days : DD
Total Present : DD
Total Absent : DD
Convenience : Rs ----

Employee page:

Employee Payroll Management System

EMPLOYEE DETAILS

	ECode	Designation	Name	Age	Gender	Email	Hr	DOB	DOJ	Proof_Id
1		Manager	Sarth Thakar	25	Male	sarth@gmail.com	scs	28-09-2000	28-09-2025	Aadhar Card
2		abc	cde	30	Female	cde@gmail.com	dcc	08-09-1995	30-09-2025	Aadhar Card

System Testing

Test Cases

Test Case ID	Test Name	Purpose	Input	Expected Output	Test Steps
TC_01	Application Launch	Verify application starts successfully.	Run Python file.	Main window opens without error.	Pass
TC_02	Add New Employee	Add a new employee record	Emp Code: 101, Name: Rahul Sharma, Designation: Manager, DOB: 12-05-1995, DOJ: 01-06-2020	Employee saved successfully in database.	Pass
TC_03	Duplicate Employee Code	Prevent duplicate employee code.	Existing Emp Code (101)	Error message "Employee already exists"	Pass
TC_04	Invalid Contact Number	Validate contact number	98765	Error message "Contact must be 10 digits"	Pass
TC_05	Age Below 21	Validate minimum age	DOB: 01-01-2010	Error message "Employee must be at least 21 years old"	Pass
TC_06	Empty Employee Code	Check mandatory employee code	Emp Code = (Blank)	Error message "Employee code is required"	Fail
TC_07	Invalid Contact Number Length	Validate contact number length	987654	Error "Contact must be 10 digits"	Fail
TC_08	Empty Employee Name	Validate mandatory name field	Name = (Blank)	Error "Name cannot be empty"	Fail
TC_09	Invalid Date Format	Validate DOB format	DOB = 32-13-2023	Error "Invalid date format"	Fail
TC_10	Experience Greater Than Age	Logical validation	Age = 25, Experience = 15	Error "Experience cannot exceed age"	Fail

Future Enhancement

1. User Authentication & Roles

- Add login system with username and password.
- Introduce role-based access (Admin, HR, Employee).
- Improve data security and accountability.

2. Web-Based Version

- Convert desktop application into a web application.
- Use Flask / Django for backend.
- Access payroll system from anywhere.

3. Advanced Reports & Analytics

- Generate monthly, yearly payroll reports.
- Export reports to PDF and Excel formats.
- Add graphs and charts for salary analysis.

4. Automated Payslip Generation

- Generate professional PDF payslips.
- Email payslips automatically to employees
- Add company logo and digital signature.

5. Bank & Payment Integration

- Integrate with bank APIs for salary transfer.
- Auto-generate bank transfer statements.
- Reduce manual payment processing.

6. Attendance Integration

- Link payroll with attendance system.
- Auto-calculate absent days from attendance.
- Improve salary accuracy.

7. Tax & Compliance Management

- Automatic income tax, PF, ESI calculations.
- Generate Form 16.
- Ensure compliance with government rules.

References

- **References:-**

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